

I acted as the lead programmer for our group and handled the overall software simulation. I coded the mathematical model for the DC motor plant and built the baseline PID control loop. I also set up the data generation pipeline to gather our training data. Using that data I designed and trained the Supervised Learning ANN (MLPRegressor). To make the project interactive I built a multithreaded Tkinter dashboard to visualize the step response and calculate our performance metrics in real time. Outside of the coding I helped write the final report. My main contributions there were writing the technical analysis of the neural network and formatting the telemetry data and comparison graphs for our final submission.